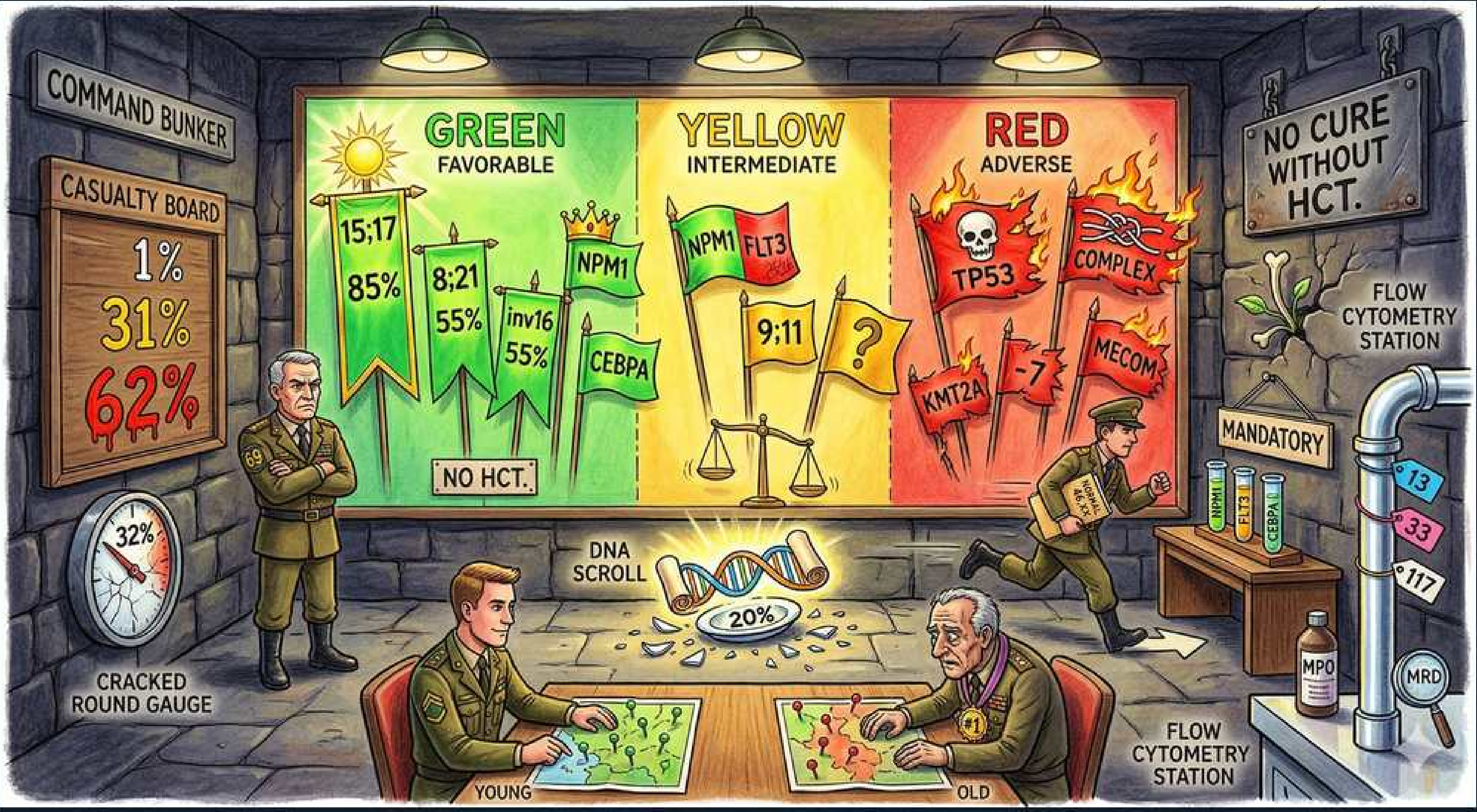
AML — Risk Stratification



1. Green = favorable

WHAT YOU SEE

The green FAVORABLE third of the wall map lit by sunshine — the green zone is the best-survival ELN tier consolidated with chemo, not CR1 transplant.

WHAT IT ENCODES

The ELN 2022 system stratifies AML into favorable, intermediate, and adverse risk using cytogenetics and molecular markers, and this drives the single biggest decision in management: whether to consolidate with chemotherapy alone or proceed to allogeneic stem cell transplant in first remission (CR1). Favorable-risk patients have the best survival and are generally consolidated with chemotherapy (e.g., high-dose cytarabine), reserving allo-HCT for relapse rather than CR1.

BOARD TRAP

Favorable-risk AML is generally NOT transplanted in CR1 — the relapse risk does not justify transplant-related mortality. Reflexively sending every AML patient to transplant is wrong; risk tier decides.

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2. t(15;17) 85%

WHAT YOU SEE

A green banner reading 15:17 with 85% — the high-percent flag is APL, the most curable AML on ATRA + arsenic.

WHAT IT ENCODES

t(15;17) creates the PML-RARA fusion of acute promyelocytic leukemia (APL/M3), the most curable AML (~85-90% cure) when treated with ATRA + arsenic trioxide (ATO), largely chemo-free in standard-risk APL. ATRA differentiates the arrested promyelocytes rather than killing them. The lethal early hazards are DIC/hemorrhage and differentiation (retinoic acid) syndrome.

BOARD TRAP

APL is treated with ATRA + arsenic, NOT 7+3 — and ATRA must be started on suspicion before genetic confirmation because of the catastrophic DIC. Watch for differentiation syndrome (fever, dyspnea, infiltrates, edema) and treat with dexamethasone.

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3. t(8;21) / inv16 55%

WHAT YOU SEE

Green banners reading 8:21 and inv16, each 55% — the paired flags are the favorable core-binding-factor leukemias.

WHAT IT ENCODES

t(8;21) (RUNX1-RUNX1T1) and inv(16)/t(16;16) (CBFB-MYH11) are the core-binding-factor (CBF) leukemias — favorable risk with good response to high-dose cytarabine consolidation. Adding gemtuzumab ozogamicin (anti-CD33 antibody-drug conjugate) further improves CBF-AML outcomes. KIT mutations can downgrade CBF prognosis.

BOARD TRAP

Core-binding-factor AML (t(8;21), inv(16)) is favorable and responds well to high-dose cytarabine — but a co-occurring KIT mutation worsens the prognosis and is the testable modifier.

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4. NPM1 / CEBPA favorable

WHAT YOU SEE

NPM1 and CEBPA flags planted in the green zone — the green-side flags are favorable NPM1 (no FLT3-ITD) and bZIP CEBPA.

WHAT IT ENCODES

NPM1 mutation WITHOUT FLT3-ITD is favorable risk, as is in-frame bZIP-mutated CEBPA. NPM1-mutant AML is also a useful MRD marker by PCR. These confer good outcomes with chemotherapy consolidation in the absence of adverse co-mutations.

BOARD TRAP

NPM1 is favorable ONLY in the absence of FLT3-ITD — NPM1 plus FLT3-ITD moves the patient into intermediate risk. The CEBPA benefit specifically requires the bZIP in-frame mutation.

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5. Yellow = intermediate

WHAT YOU SEE

The yellow INTERMEDIATE middle panel with a balance scale — the teetering scales are intermediate risk, where CR1 transplant is generally advised.

WHAT IT ENCODES

Intermediate-risk AML (including normal karyotype not otherwise classified, NPM1 with FLT3-ITD, and t(9;11)/KMT2A-MLLT3) has an outcome poor enough that allogeneic transplant in CR1 is generally recommended if a suitable donor exists and the patient is fit. MRD status increasingly refines this decision.

BOARD TRAP

Intermediate-risk patients are typically referred for allo-HCT in CR1, unlike favorable-risk patients. The transplant decision hinges on risk tier plus fitness and donor availability, not on age alone.

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6. NPM1+FLT3 / t(9;11)

WHAT YOU SEE

Yellow-zone flags reading NPM1, FLT3, and 9:11 with a question mark — the yellow flags are NPM1+FLT3-ITD and t(9;11), the intermediate combos.

WHAT IT ENCODES

NPM1-mutant AML co-occurring with FLT3-ITD falls into intermediate risk (the FLT3-ITD pulls the otherwise-favorable NPM1 down), and t(9;11)(p21;q23)/KMT2A-MLLT3 is also intermediate. FLT3-mutated patients additionally receive midostaurin added to induction/consolidation.

BOARD TRAP

The combination NPM1mut + FLT3-ITD = intermediate, whereas NPM1mut alone = favorable — the FLT3-ITD is the deciding factor. Note t(9;11) is the one KMT2A rearrangement that is intermediate, while other KMT2A rearrangements are adverse.

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7. Red = adverse

WHAT YOU SEE

The red ADVERSE panel engulfed in flames — the burning red zone is the worst-survival tier pushing hard toward CR1 allo-HCT.

WHAT IT ENCODES

Adverse-risk AML has the worst survival and the highest relapse rate, so allogeneic transplant in CR1 is strongly favored in any fit patient with a donor. This category includes TP53, complex/monosomal karyotype, -7/del(7q), -5/del(5q), KMT2A rearrangements (other than t(9;11)), MECOM(EVI1), RUNX1, ASXL1, and BCR-ABL1.

BOARD TRAP

Adverse-risk AML pushes hard toward allo-HCT in CR1, but TP53-mutated/complex-karyotype disease responds poorly even to transplant — outcomes remain dismal, which is itself frequently tested.

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8. TP53 / complex

WHAT YOU SEE

Red flags reading TP53 and COMPLEX in the flames — the fiery flags are TP53 and complex karyotype, chemoresistant even after transplant.

WHAT IT ENCODES

TP53 mutation and complex karyotype (≥3 abnormalities) or monosomal karyotype define the worst-prognosis AML, with chemoresistance and very low cure rates even with transplant. TP53-mutant AML is often therapy-related/secondary and responds poorly to both intensive chemo and HMA+venetoclax.

BOARD TRAP

TP53 mutation is the single most ominous AML lesion — it predicts resistance to standard chemo AND to HMA/venetoclax, with poor transplant outcomes; clinical-trial enrollment is often preferred. TP53 also overlaps heavily with therapy-related AML.

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9. KMT2A / -7 / MECOM

WHAT YOU SEE

Red flags reading KMT2A, -7, and MECOM in the adverse zone — the flaming flags are the adverse cytogenetic lesions (KMT2A except t(9;11)).

WHAT IT ENCODES

KMT2A (MLL, 11q23) rearrangements other than t(9;11), monosomy 7 (-7)/del(7q), and MECOM(EVI1) rearrangements [inv(3)/t(3;3)] are adverse-risk. -7 and complex karyotypes are classic for alkylator-induced therapy-related AML, and KMT2A/11q23 for topoisomerase-II-inhibitor-induced t-AML.

BOARD TRAP

KMT2A rearrangements are adverse EXCEPT t(9;11), which is intermediate — the exception is the trap. Monosomy 7 and KMT2A also flag secondary/therapy-related AML, linking cytogenetics back to etiology.

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10. Survival 1/31/62%

WHAT YOU SEE

A casualty board posting 1%, 31%, 62% mortality tiers — the stark numbers show survival diverging across the three ELN risk groups.

WHAT IT ENCODES

Outcomes diverge dramatically across ELN tiers — favorable APL/CBF leukemias approach high cure rates, while adverse-risk TP53/complex disease has very low long-term survival. This gradient is precisely why genetic risk classification is mandatory before choosing consolidation strategy.

BOARD TRAP

Prognosis in AML is driven primarily by genetics/cytogenetics, not by the morphologic FAB subtype — the old FAB classification has been supplanted by molecular ELN risk grouping.

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11. ≥20% blasts / DNA scroll

WHAT YOU SEE

An unrolled DNA scroll stamped 20% on the table — the scroll is the ≥20% blast diagnostic threshold (waived for AML-defining genetics).

WHAT IT ENCODES

AML is diagnosed by ≥20% myeloblasts in blood or marrow, OR by an AML-defining recurrent genetic abnormality regardless of blast count. The defining genetics that diagnose AML below 20% blasts include t(15;17)/PML-RARA, t(8;21), inv(16)/t(16;16), and other recurrent fusions; an extramedullary myeloid sarcoma also qualifies.

BOARD TRAP

AML-defining genetics (APL, CBF, etc.) establish the diagnosis even when blasts are <20% — the 20% threshold is not absolute. Note: with TP53/MDS-related changes, WHO/ICC now allow an AML or MDS/AML diagnosis at 10-19% blasts.

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12. Young vs old

WHAT YOU SEE

A young soldier and an old soldier seated across the planning table — the two ages are fitness modifying therapy intensity without overriding genetic tier.

WHAT IT ENCODES

Age, performance status, and comorbidities determine fitness for intensive therapy and transplant eligibility, which interacts with genetic risk. Older or unfit patients receive HMA + venetoclax rather than 7+3 regardless of tier, and transplant is offered to fit intermediate/adverse-risk patients with an acceptable donor.

BOARD TRAP

Age and fitness modulate the therapy intensity and transplant decision but do NOT override the genetic risk tier — an old fit patient can still be transplanted, and a young patient with favorable APL still avoids CR1 transplant.

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13. MPO flow cytometry

WHAT YOU SEE

A flow-cytometry station racked with tubes labeled MPO and MRD — the lab bench is MPO/immunophenotyping that confirms myeloid lineage and tracks MRD.

WHAT IT ENCODES

Myeloperoxidase (MPO) positivity confirms myeloid lineage and distinguishes AML from ALL, and flow cytometry establishes the immunophenotype (myeloid blasts express CD13, CD33, CD117, and CD34; monocytic markers CD14/CD64). Flow and molecular MRD assessment after induction are mandatory and increasingly guide post-remission transplant decisions.

BOARD TRAP

MPO positivity (cytochemistry or flow) distinguishes AML from ALL — ALL blasts are MPO-negative. CD33 expression also identifies candidates for gemtuzumab ozogamicin and CD34/CD117 mark myeloid blasts.
